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REPUBLIC OF SOUTH AFRICA

Life and living

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
2½ weeks	Plants and animals on Earth	**Many different plants and animals** • there are many different plants and animals living in different habitats on Earth* (South Africa has a wide variety of indigenous plants and animals and their habitats) **Inter-dependence** • plants and animals depend on each other • they also depend on the resources available (such as air, water, soil, food, and places to hide) in their own habitats **Animal types** • there are many different kinds of animals, - some do not have bones, and some have hard outer 'skins' or shells (invertebrates) • some have bones (vertebrates)	• Pictures of plants and animals	• Identify different habitats in South Africa and some of the plants and animals that we find there • Describe and compare animals without bones to animals with bones • Describe interdependence between living and non-living things • Identify the interdependence between the animals and/or plants and the non-living things in their environment • Identify common characteristics of invertebrates and vertebrates
1 week	Animal skeletons	**Skeletons of vertebrates** • a vertebrate skeleton consists of bones and joints, and is inside the body • bones are hard and form a strong frame structure • a skeleton provides support for an animal's body and protection for its organs; - skull - protects the brain - backbone with vertebrae - protects the spinal cord - ribs - protect the lungs and heart - shoulder blades, arms, legs, hip bones – for movement **Movement** • vertebrate animals can move because there are - muscles attached to the skeleton • joints between the bones*** (*In this grade, details about how muscles are attached, and the structure of joints are not required*)	• Pictures and examples of animal skeletons/bones	• Identify the different types of skeletons • Use pictures of animals to identify the five groups of vertebrates and their common characteristics • Identify and describe different bones in a vertebrate skeleton and state the functions of each bone • Label a diagram of the human skeleton • Describe how different vertebrate animals move, including humans • Design, draw, make and evaluate a skeleton. Write a paragraph about the skeleton that you built to address what worked and what did not work. Your skeleton should have the following specifications: It must be three-dimensional, it must look realistic, it must have/show the basic parts, i.e. skull, backbone and ribs, and it must be strong and rigid so that it can stand on its own
2 weeks	Skeletons as structures	**Frame and shell structures** • a vertebrate skeleton is a frame structure (*also refer to grade 4 Matter & Materials*) • some invertebrate skeletons are shell structures such as that of a crab	• Paper, drinking straws, wooden dowels or sticks (30 cm × 10 mm), sticky tape and metal paper fasteners	
1½ weeks	Food chains	**Food and feeding** • green plants make their own food** and build their branches and stems using water and carbon dioxide from the air, and energy from sunlight. (*This important process is called "photosynthesis." However no further detail is required in this grade, learners will deal with it in detail only in higher grades*). Plants use carbon dioxide from the air and release oxygen into the air • animals need food to carry out their life processes (to move, feed, grow, sense the environment, excrete, breathe and reproduce) • all animals depend on plants as their primary source of food (herbivores, carnivores and omnivores) • a food chain describes the feeding relationships between plants and animals. • a food chain - starts with a plant, (produces foods) - then follows with an animal that eats the plant - after that with an animal that eats that animal • includes the transfer of energy which flows from the plant through to the last animal in the chain	• Pictures of various plants and animals	• Describe how each living thing gets food and how energy is passed from one organism to the next • Sequence plants and animals to make up a proper food chain in which the energy is transferred from one organism to the next, with up to four organisms each, and describe their relationships • Draw a food chain using common plants and animals with at least one sentence next to, below, or above each arrow. • Classify animals according to their feeding relationships (as herbivores, omnivores, carnivores, scavengers or decomposers) • Identify predators and their prey • Explain the four stages in the life cycle of a

2 weeks	**Life cycles**	**Growth and development** • plants and animals grow and develop throughout their lives • a life cycle describes the stages and processes that take place as a plant or animal grows and develops • a life cycle describes how one generation of a plant or animal gives rise to the next generation through reproduction • death can occur at any stage of the life cycle • many animals care for their young in order for them to grow and develop	• Pictures of different stages in the development of various plants and animals	flowering plant • Describe the different stages in the life cycle of an animal
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Practical task/investigation • Test			

Matter and materials

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
2½ weeks	Metals and non-metals	**Properties of metals** • metals are used to make things because they have certain properties • some properties of metals - shiny - hard - strong - can be hammered, shaped (malleable) and made into thin wires without breaking (ductile) - melt at high temperatures • metals are mined from the Earth* (*How and where metals are mined is covered in Social Sciences*) **Properties of non-metals** • non-metals are used to make things because they have certain properties • some properties** (*There are many different non-metals and they have a variety of different properties. Here the focus is only on two of the most general properties*) of solid non-metals - dull - can break easily (brittle)	• Examples of metal objects such as copper wire, coins, nails, cooking pots or knives and forks • Examples of non-metal objects such as a piece of chalk, a pile of sand or a piece of coal	• Revision: Grade 4: States of matter • Investigate, compare and record the properties of some metal objects (such as copper wire, coins, nails, cooking pots, knives and forks) and some non-metal objects (such as a piece of chalk, a stone, a pile of sand or a piece of coal) • Identify properties of metals (i.e., conducting electricity and conducting heat) and relate these properties to uses of metals. • Investigate ways to make old and dull metal objects shiny again • Investigate how rust occurs • Research and write about the properties and uses of metals from the home environment
2½ weeks	Uses of metals	**Other properties of metals** • metals are useful because of their special properties - metals conduct heat - some metals are magnetic and some are not - only iron rusts (some metals tarnish or become dull) **Uses of metals** • metals are used to make things such as coins, wire, jewellery, furniture, buildings and bridges, motor cars, kitchen utensils, roofs	• Magnets and objects such as coins, iron filings, nails, drawing pins, paper clips or wire	
2½ weeks	Processing materials	**Combining materials** • materials can be processed to make new materials/products, such as - mixing and setting (such as plaster of Paris (or Polyfilla) and water to make plaster) - mixing and setting (such as sand, gravel, cement and water to make concrete) - mixing (such as flour and water to make a sticky paste that can be used as glue) - mixing and cooking (such as making dough) - mixing and cooling (such as jelly powder and water to make jelly) - mixing, drying and/or firing (such as wet clay and straw to make clay bricks) • the properties of the new materials/products may be different from the properties of the materials we started with	• Materials and substances such as plaster of Paris (or Polyfilla), sand, gravel, cement, flour, ingredients to make dough, jelly powder, wet clay and straw	• Investigate reasons why we process materials • Describe, with examples, the properties of processed materials • Explain, with examples, the purpose of processing materials • Explain the difference between raw materials, natural materials and processed materials • Research the traditional processing methods that humans have been using to give materials more desirable properties
1½ weeks	Processed materials	**Properties and uses** • materials such as plaster of Paris, concrete, fabrics, ceramics and glass, plastics and paints, have their own special properties • processed materials are useful because of their special properties. They can be strong, durable, waterproof, fire resistant, have interesting colours or textures	• Clay • Pictures and examples of objects made by weaving plant material	
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Practical task/investigation • Test			

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Energy and change

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
3 weeks	**Stored energy in fuels**	**Fuels** • energy is stored in fuels (including food*) • we use fuels as sources of useful energy • everyday fuels that we use include coal, wood, petrol, paraffin, gas and candle wax • when we burn these fuels we get useful output energy such as heat and light **Burning fuels** • fuels need heat to set them alight, and air (oxygen) to keep on burning **Safety with fire** • fires can be a threat in our communities	• Examples of substances including wood, coal, candle (wax), paraffin, peanut, a biscuit • Candles and different sized glass containers • Cells (batteries), lengths of wire, light bulbs	• Compare energy from various packaging for foods collected from home • Examine various fuels including wood, coal, candle (wax), paraffin, peanut, a biscuit. Burn three different fuels from above, and compare and describe the • input energy needed to make them burn • output energy obtained from the fuel • Investigate how long a candle will burn when covered with different sized glass containers (the candles will stop burning when all the oxygen is used up) • Write and draw about fires in our communities including causes, prevention and act out what action to take during a fire [This can be used as a possible project]
3 weeks	**Energy and electricity**	**Cells and batteries** • energy can be stored in torch cells and batteries • a circuit is a system that transfers electrical energy to where it is needed **Mains electricity** • electricity from the power station is transferred in a circuit to our homes and back to the power station • a power station needs a source of energy • the source of energy in a power station can be a fuel such as coal** (*The focus here is on coal-fired power stations. There are also other power stations such as hydro-electric and nuclear power stations*) **Safety with electricity** • safety precautions should be taken when using electricity	• Cells (batteries), lengths of wire and light bulbs	• Investigate the source of electricity in a torch • Compare the differences between batteries and cells • Explore and explain various ways of making a complete simple circuit • Draw simple circuit diagrams with correct symbols and labels • Use diagrams to trace and explain how the electricity comes from the power station to our homes/schools, including the power station, pylons, substations, electricity boxes, wall sockets, plugs and appliances such as a TV, kettle, stove, torch, radio, iron, fan/hair dryer and computer, etc. • Use pictures and illustrations to explain safety tips for using electricity
2 weeks	**Energy and movement**	**Elastic and springs** • we can make things move using stretched or twisted elastic and compressed springs • when we stretch or twist elastic or compress a spring, we store energy in it • when we release the elastic or spring again, we get movement energy	• Elastic bands and compressed springs, a catapult, elastic powered aeroplanes, “jack-in-a-box”	• Explain how stored energy can be changed into movement energy using elastic bands, a compressed metal spring, etc. • Investigate the different ways in which stored energy can be changed into movement energy using elastic bands, a compressed metal spring, etc.
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Practical task/investigation • Test			

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Planet Earth and Beyond

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
1½ weeks	Planet Earth	**The Earth moves** • the Earth travels in an orbit (pathway) around the Sun • the Earth takes about 365 days to travel once around the Sun, this is called a year • the Earth spins on its own axis • the Earth takes about 24 hours to spin once, this is called a day	• Pictures of the Earth showing its main features • Video clips • Pictures and models of the Earth, Moon, Sun and other planets • A light source such as a torch, lamp/candle	• Describe and identify the main features of the Earth • Draw or make models of the Earth • Demonstrate the Earth's movement in its orbit around the Sun • Describe the Earth's movement around its own axis • Identify the main elements (soil, air, water and sunlight) that support life on Earth • Identify and describe different soil types correctly • Explain the formation of sedimentary rock • Distinguish between body and trace fossils • Explain aspects of South Africa's fossil record
2 weeks	The surface of the Earth	**Rocks** • the surface of the Earth is called the crust, and consists of rocks (even under the oceans), and soil • soil, air, water and sunlight support life on Earth **Soil comes from rocks** • the land is made up of rocks, subsoil and topsoil • soil supports life on Earth • topsoil lies on the surface – topsoil is formed when rocks break into small grains over time **Soil types** • soil is usually a mixture of different types of soil grains in different proportions - sandy soil – has a high proportion of course sand grains - clayey soil – has a high proportion of fine grains of clay - loamy soil – has a mixture of sand, clay and other soil grains. Loamy soil also contains humus (decomposed compost) • the soil also has air, water, remains of dead organisms and very small living organisms in it • soil forms very slowly in nature – once topsoil is lost, it cannot be replaced, and thus we need to conserve it	• Samples of different types of soil • Measuring cylinders, funnels and filter paper and beakers • Seeds and rulers to measure length • Sandy soil • Clayey soil • Loamy soil • Stones • Dry plant material	
2 weeks	Sedimentary rocks	**Formation of sedimentary rock** • sedimentary rocks are formed over a very long time in the following way: - first, rocks break down into small grains - then, mud and sand is moved by wind and water - after that, mud and sand gets deposited in low lying areas - over time, new layers of mud and sand are deposited on top of existing layers - after a very long time, these layers become compacted and hardened and form sedimentary rock • sedimentary rocks always have visible layers within the rock • examples of sedimentary rock are shale, sandstone and limestone **Uses of sedimentary rock** • limestone is used to make cement • sandstone and shale are used in buildings	• Pictures and/or samples of sedimentary rock such as limestone and sandstone	
2½ weeks	Fossils	**Fossils in rock** • fossils are the remains of ancient plants and animals preserved in rock - fossils are found in some layers of sedimentary rock - fossils are evidence/a record of the history of life on Earth • there are two main types of fossils; body and trace fossils **Body and trace fossils** • body fossils form from the hard parts of plant and animal bodies including teeth, bones, shells, stems, leaves and seeds	• Pictures and/or samples of sedimentary rock • Play dough, clay, plaster of Paris and a variety of parts of plants and animals • Pictures of fossils • Information texts about South African fossils	

		• trace fossils form from traces left by animals including footprints, nests, eggs and droppings • some features of fossils resemble the features of plants and animals living today **Importance of South African fossils** • South Africa has a particularly rich fossil record of plants, animals and early humans • important fossils found in South Africa include the Coelacanth and African dinosaurs • the "*Cradle of humankind*" is one of the sites where important fossils of humankind have been found in South Africa		
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Test			

Major Process and Design Skills

The teaching and learning of Natural Sciences and Technology involves the development of a range of process and design skills that may be used in everyday life in the community and in the workplace. Learners also develop the ability to think objectively and use a variety of forms of reasoning while they use these skills. Learners can gain these skills in an environment that taps into their curiosity about the world, and that supports creativity, responsibility and growing confidence.

The following are the cognitive and practical process and design skills that learners will be able to develop in Natural Sciences and Technology.

1. ***Accessing and recalling information*** – being able to use a variety of sources to acquire information, and to remember relevant facts and key ideas, and to build a conceptual framework
2. ***Observing*** – noting objects, organisms and events in detail
3. ***Comparing*** – noting similarities and differences between things
4. ***Measuring*** – using measuring instruments such as rulers, thermometers, clocks and syringes (for volume)
5. ***Sorting and classifying*** – applying criteria to sort items into a table, mind-map, key, list or other format
6. ***Identifying problems and issues*** – being able to articulate the needs and wants of people in society
7. ***Raising questions*** – being able to think of and articulate relevant questions about problems, issues and natural phenomena
8. ***Predicting*** – stating, before an investigation, what the learner thinks the results will be for that investigation
9. ***Hypothesising*** – putting forward a suggestion or possible explanation to account for certain facts. A hypothesis is used as a basis for further investigation that will prove or disprove the hypothesis
10. ***Planning investigations*** – thinking through the method for an activity or investigation in advance. Identifying the need to make an investigation a fair test by keeping some things (variables) the same, while other things will vary
11. ***Doing investigations*** – this involves carrying out methods using appropriate apparatus and equipment, and collecting data by observing and comparing, measuring and estimating, sequencing, or sorting and classifying. Sometimes an investigation has to be repeated to verify the results
12. ***Recording information*** – recording data from an investigation in a systematic way, which includes drawings, descriptions, tables and graphs
13. ***Interpreting information*** – explaining what the results of an activity or investigation mean (this includes reading skills)
14. ***Designing*** – showing (e.g. by drawing) how something is to be made, taking into account the design brief, specifications and constraints
15. ***Making/constructing*** – building or assembling an object using appropriate materials and tools and using skills such as measuring, cutting, folding, rolling and gluing
16. ***Evaluating and improving products*** – using criteria to assess a constructed object and then stating or carrying out ways to refine that object
17. ***Communicating*** – using written, oral, visual, graphic and other forms of communication to make information available to other people.